High-Performance Flip-Chip BGA Packages Reliability Prediction in Automotive Applications

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***Abstract*—Recently, the demand of IC Packages has grown rapidly in automotive applications. Advanced Driver Assistance Systems (ADAS) require more powerful and more chips to enable Level 2 and higher systems. However, automotive package operates in severer environment than consumer package, so it requires higher reliability whether in package-level or board-level reliability tests. To ensure packages reliability, a more comprehensive approach needs to be implemented. In this study, our aim is to predict high-performance flip-chip BGA (HFCBGA) substrate copper trace fatigue life in temperature cyclic test (TCT). HFCBGA is the composite package of FCBGA with heat spreader that is used to extend the heat conduction area and enhance package warpage control because of its high power and high speed demand. We used advanced Metrology Analyzer (aMA) to measure package warpage especially from TCT condition -55°C to 125°C and then used finite element method (FEM) to establish 3-D numerical model for validation. Afterwards, we built up a detailed sub-model to simulate substrate copper trace plastic work density (PWD) considering Anand's model under thermal cycling, and brought the PWD increment into Coffin-Manson Model to predict substrate copper trace fatigue life. From the simulation results, we can get the key factors affecting substrate copper trace fatigue life. Through different structure designs and BOM selection such as heat spreader designs, adhesive types, soldermask types, trace layout, etc., substrate copper trace broken risk can be lowest to meet automotive specification of AEC - Q100.**

Keywords*—*Automotive, HFCBGA, TCT, aMA, FEM

I. Introduction

Automotive, AI, 5G, wearable devices, AR, VR and other industries are fast growing, caused the IC has gradually become high performance, high reliability, denser and thinner. With the circuit continuous miniaturization, the package has also become smaller. In Automotive, Advanced Driver Assistance Systems (ADAS) are more powerful and help humans drive more safe and convenient, but more chips are needed to meet the requirements of Level 2 and higher systems. In high performance package solutions, High-Performance Flip-Chip Ball Grid Array (HFCBGA) is one of them and is the research target of this paper. HFCBGA (Fig. 1) products have been in high demand of server, graphics, high-end application, etc. Ian Hu et al. [1] were studied HFCBGA thermal and reliability performance. HFCBGA connects die and substrate through C4 bump or copper pillar bump, Robert Darveaux [2] and Meng-Kai Shih et al. [3] were used FEM to predict solder fatigue life. Next, using CUF (Capillary Underfill) process to fill die and substrate gap. Then heat spreader attaches on substrate through DAF (Die Attach Film), between die and heat spreader gap, TIM (Thermal Interface Material) is used to fill the gap and conducts the heat from die to heat spreader. Tong Hong Wan et al. [4] were studied HFCBGA warpage performance for different temperature profile. In order to confirm the reliability of the product, the product is usually conducting Thermal Cyclic Test (TCT). Through the TCT, it can be known how many cycles the product will fail and whether it meets the product specification. In the experiment, product will be placed in the oven for TCT, and the product will be taken out every 250 or 500 cycles to measure the resistance. If there is any abnormality, it will need to do failure analysis to find out failure mode. Generally, conduct experiments can know each factor effect on product and then optimize the product design, but the cost of the experiment is too high. In order to confirm product design in the early stage and know the effect of various factors on product, simulation can not only understand the relationship between different structures and BOM, but also have advantages in cost and time. It can be also found out the risk location of product and improve. In this study, the copper trace fatigue life is established by FEM and experiments, and then improve the copper trace fatigue life through different structures and BOM to meet the AEC - Q100 specification.

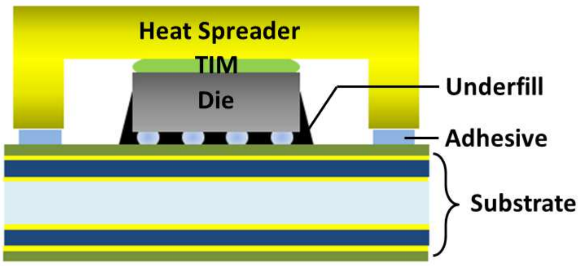

Fig. 1. HFCBGA SCHEMATIC

II. Test Vehicle

The package body size used in this work is 23x23 mm^2. Die size is 13x11 mm^2, substrate and heat spreader thickness are 0.7 mm and 0.8 mm respectively. In order to meet the AEC - Q100 Grade 2 specification [5], samples need to be placed under TC-B condition (-55°C to 125°C) shown as Fig. 2 for 1000 cycles test and pass O/S test.

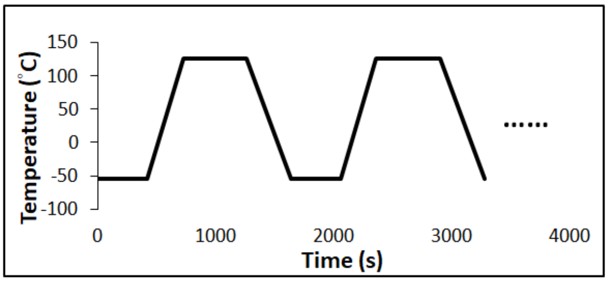

Fig. 2. TC-B CONDITION PROFILE

III. Finite Element Method

In this study, we use ANSYS to build FEM model and conduct simulation. In the simulation settings, all components are perfectly bonded. The boundary conditions of the model are shown as Fig. 3. The sub-model is established as Fig. 4. After calculating global model, sub-model is located at crack area and output the local deformation calculated by the global model into the sub-model as boundary conditions. This method is used as a standard to simulate different legs.

Material properties are shown as Table I. All materials are homogeneous and isotropic. In order to simulate copper trace fatigue life, copper viscoplastic behavior need to be considered to calculate Plastic Work Density (PWD) by following Anand's model constitutive equation:

$$\frac{d\varepsilon_p}{dt} = A\left[\sinh(\varepsilon\sigma/s)\right]^{1/m}\exp\left(-\frac{Q}{kT}\right) \quad (1)$$

Which is widely used in the literature [6] and the parameters are shown as Table II. The larger the PWD, the shorter the failure cycle, while the smaller the PWD, the longer the failure cycle, respectively. After calculating copper PWD, use Coffin-Manson model to calculate copper fatigue life:

$$Nf = A \times \Delta W^{-B} \quad (2)$$

Where A is a constant and B is larger than 1.4 based on in-house experiment obtained by curve fitting (Fig. 5), ΔW is the PWD increment between cycles in the copper trace fatigue region.

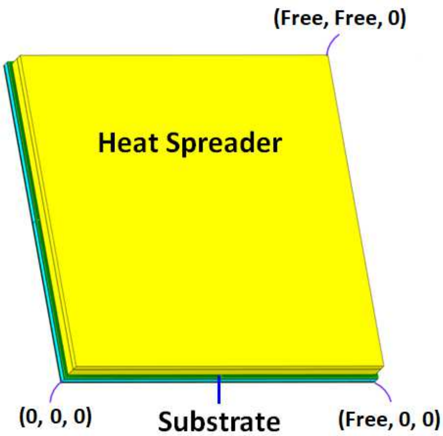

Fig. 3. GlOBAL MODEL BOUNDARY CONDITIONS

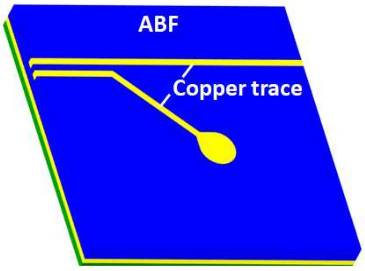

Fig. 4. SUB-MODEL SCHEMATIC

TABLE I. MATERIAL PROPERTY

Material	RT/HT Modulus (GPa)	CTE (ppm/°C)	T_g (°C)
Die	131	2.8	-
Copper	117	16.9	-
Core	26/17	5/1	260
ABF-1	5/0.1	40/120	150
ABF-2	8/0.1	20/80	155
Underfill-1	7/0.07	30/130	100
Underfill-2	6/0.05	31/129	95
TIM	0.04	235	-
Soldermask-1	5/0.2	40/130	130
Soldermask-2	3/0.3	35/120	130
Adhesive-1	0.1	120	-
Adhesive-2	0.004	250	-

TABLE II. ANAND'S MODEL OF COPPER FOIL

Parameter	Value
S_0 (MPa)	0.001
Q/k (1/K)	6124.7
A (1/sec)	12000
ξ	2.807
m	0.03
h_0 (MPa)	96100
s^ (MPa)	1317
n	0.00001
a	1.001

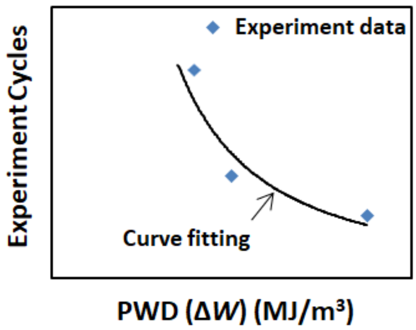

Fig. 5. CURVE FITTING OF COFFIN-MANSON MODEL

IV. Results and Discussion

A. advanced Metrology Analyzer and Validation

In order to verify FEM model, we use advanced Metrology Analyzer (aMA) to measure package warpage. Fig. 6 shows the aMA machine that contains heating chamber, two high resolution charge coupled device (CCD) cameras for image capture on specimen surface and motorized linear stages. The specimen needs surface pretreatment to generate random speckles and is measured from room temperature to peak reflow temperature and then return back to room temperature (25°C-260°C-25°C). However, we especially measure package warpage at -55°C in this study because TC-B condition is -55°C to 125°C. aMA can produce thermal deformation images of out-of plane and in-plane during different temperatures. From Fig. 7, warpage is measured by bottom side substrate surface and 92.5 µm warpage of convex shape at -55°C. The simulation result is 90.3 µm of convex shape at -55°C, which is match with measurement result, and the gap is within 5%.

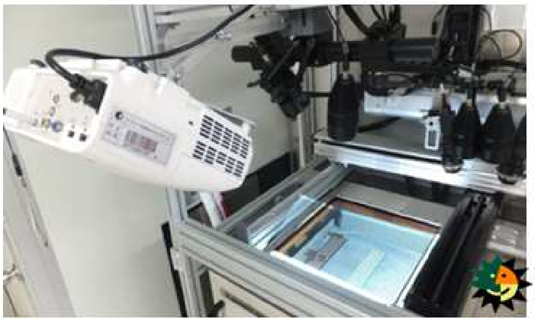

Fig. 6. ADVANCED METROLOGY ANALYZER

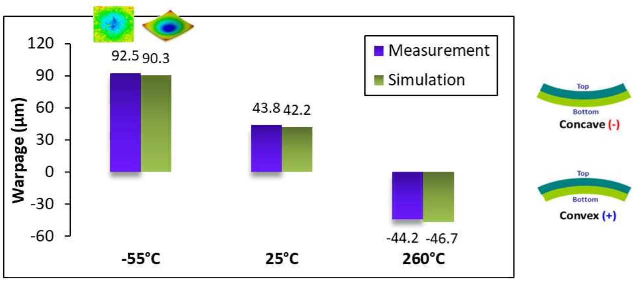

Fig. 7. WARPAGE VERIFICATION

B. Mechanism Discussion

In the previous chapter, it can be seen that package has maximum warpage at -55°C because of maximum temperature gradient from stress free temperature to -55°C. From heat spreader side and substrate side CTE mismatch, package has thermal-induced warpage and substrate withstands cyclic bending through -55°C to 125°C TCT, causing substrate copper trace crack risk shown as Fig. 8. From simulation result, substrate copper layer maximum stress is located between die corner and heat spreader (Fig. 9 red color area), we further conduct simulation of sub-model to calculate copper trace PWD, maximum PWD is located near tear, shown as Fig. 10. To investigate the bonded force between heat spreader and substrate, we also measure the adhesive Young's modulus from -55°C to 260°C shown as Fig. 11. It can be observed that adhesive-1 modulus is ramp up rapidly near -40°C that is adhesive-1 glass transition temperature (T_g). Substrate copper trace withstand higher stress in -55°C because higher adhesive modulus result in higher bonded force between substrate and heat spreader, eventually causing substrate copper trace crack in TCT.

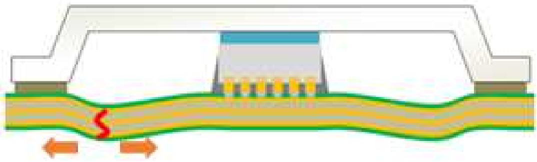

Fig. 8. SUBSTRATE COPPER TRACE CRACK

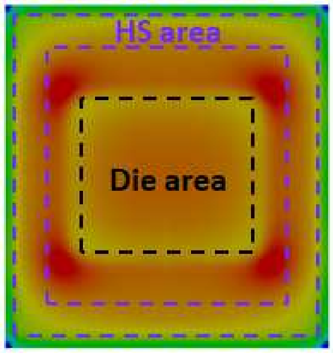

Fig. 9. COPPER LAYER STRESS CONTOUR

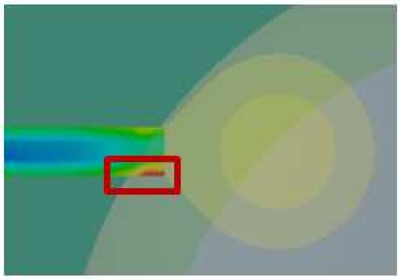

Fig. 10. MAXIMUM COPPER TRACE PWD

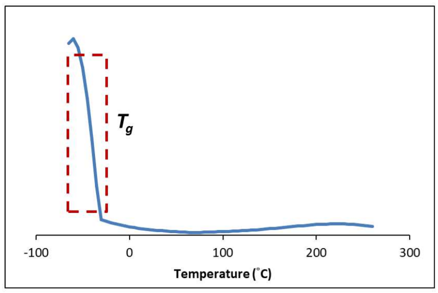

Fig. 11. ADHESIVE-1 YOUNG'S MODULUS DATA

C. BOM Comparison Simulation

To pass 1000 cycles of TC-B condition, we conduct simulations for different BOM including ABF, Soldermask, Adhesive and Underfill types, and we can get the key material affecting substrate copper trace fatigue life. From the prediction results shown as Fig. 12, ABF-1 shows better performance because of lower modulus. Soldermask-2 has lower CTE and modulus and shows 1.05 times life. However, adhesive-2 shows 1.12 times life, indicating that adhesive has the significant effect to copper trace life. Since heat spreader constraints the substrate deformation through adhesive, bonded force between heat spreader and substrate is stronger when adhesive has higher modulus, causing lower substrate copper trace fatigue life. From Underfill-2 simulation result, it shows insignificant effect. Therefore, adhesive and soldermask show more significant effect on substrate copper trace fatigue life.

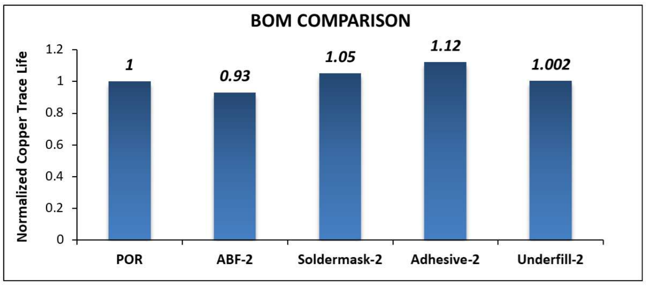

Fig. 12. BOM COMPARISON

D. *Structure Comparison Simulation*

After BOM comparison, we modify package structure to know the structure index affecting substrate copper trace life. From the prediction results shown as Fig. 13, thicker (5 µm) soldermask and copper shows 1.06 times life and also enhance package strength. In order to reduce the bonded force between heat spreader and substrate, adhesive pattern is modified from 2U to 4I pattern, shown as Fig. 14, it shows 1.02 times life. Additionally, we modify heat spreader type to Direct Lid Attach (DLA), which can reduce bonded force between heat spreader and substrate by removing the lid foot, shown as Fig. 15, it shows 1.18 times life.

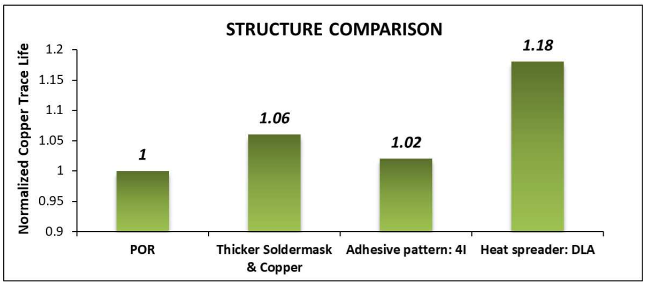

Fig. 13. STRUCTURE COMPARISON

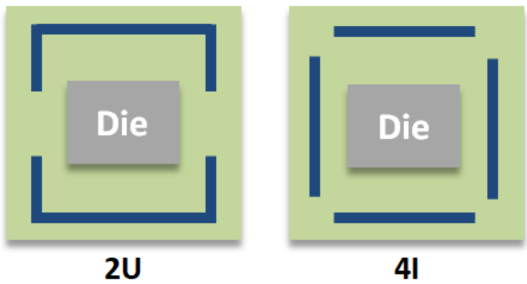

Fig. 14. ADHESIVE PATTERN

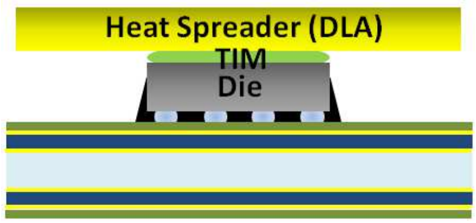

Fig. 15. HEAT SPREADER: DLA

V. CONCLUSION

In this study, we establish 3D-numerical model, and use Anand's model and Coffin-Manson model to build a prediction model for substrate copper trace fatigue life. By aMA measurement, we can obtain the package warpage from -55°C to 260°C and validation with simulation results, it shows well-validated and gap is within 5%. From the mechanism study, we know substrate copper trace crack due to TCT cyclic bending, and simulation result indicates that the maximum copper layer stress is located between die corner and heat spreader, near tear of copper trace. We also measure the adhesive-1 modulus, and find adhesive-1 T_g is near -40°C, so the modulus is ramp up rapidly in this temperature range, causing higher heat spreader and substrate bonded force and higher copper trace crack risk. From BOM comparison simulation, we realize that substrate copper trace shows better prediction life when ABF-1, soldermask-2 and adhesive-2 have lower modulus. From structure comparison simulation, thicker soldermask and copper, adhesive pattern from 2U to 4I and DLA type heat spreader can enhance substrate copper trace fatigue life. This approach enables quickly prediction of substrate copper trace fatigue life and effectively reduces experiment cost.

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